

Kevin Rodriguez

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EDUCATION

University of Central Florida

Bachelors of Science in Computer Science

Organizations - Knight Hacks, IEEE of UCF, SHPE of UCF

Orlando, FL

Jan. 2027

TECHNICAL SKILLS

Languages: C++, Python, Java, C, JavaScript, TypeScript, Go

Frameworks & Libraries: FastAPI, Express.js, React, Tailwind CSS, OpenCV, MediaPipe, OSNet, DeepStream

Databases: MySQL, MongoDB, Firebase, Supabase, PostgreSQL, NeonDB

Cloud, DevOps & Tools: AWS, Docker, Git, Databricks, Jira

EXPERIENCE

Incoming Software Engineer Intern – TIP Program

Capital One

Jun. 2026 - Aug. 2026

Chicago, IL

Center Director of Education

Mathnasium

Nov. 2024 – Present

Orlando, FL

- Oversee academic progress for **200+ students** by developing personalized learning plans, resulting in a **15% average improvement in assessment scores**.
- **Lead and mentor a team of 15+ instructors**, conducting **training sessions** and **performance reviews** to ensure **consistent** instructional **quality** across all levels.

PROJECTS

Cyclops | *YOLOv11, DeepSORT, Gemini API, FireBase*

- Engineered an AI-driven backpack-mounted safety system that achieved **90% accuracy** in detecting potential threats using YOLOv11 and DeepSORT for **real-time object tracking and re-identification**.
- Integrated Firebase Cloud Messaging with an Android app to deliver **sub-2-second alerts** when suspicious activity was detected in the user's blind spot
- Enhanced environmental awareness by fusing **ultrasonic sensor data** (Arduino) with CV outputs, improving detection confidence by **20%**.

Crisis Averted | *FastAPI, Databricks, ActianDB, Vultr, Elevenlabs, Three.js*

- **Won 3rd Place Overall** out of **1000+ hackers**.
- Led system architecture and backend development of an AI-driven humanitarian intelligence platform, designing a distributed analytics pipeline across Databricks, ActianDB, and GPU infrastructure on Vultr to deliver **real-time semantic search** and funding gap analysis with **sub-2-second query performance**.
- Ingested, cleaned, and normalized **140k+ records of CSV datasets** in Databricks, building structured schemas and transformation pipelines to support **cross-country funding gap analysis from 2022–2026**.

PlayedIt | *MERN Stack, IGDB API, Machine Learning, REST API*

- Designed and implemented **30+ RESTful API** endpoints for authentication, profiles, and recommendations, reducing backend response times by **40%** through optimized queries and indexing.
- Engineered a scalable MongoDB data model and optimized retrieval via **Mongoose aggregation pipelines and lean queries**, cutting database load by **30%** during high-traffic operations.
- Integrated IGDB game metadata with custom API logic supporting search, pagination, and dynamic cover fetching—**enabling 50,000+ games** to be browsed smoothly.

Pipeline to Better Placemaking | *DeepStream 9.0, MV3DT, TensorRT, PeopleNET, GStreamer*

- Replaced manual homography and sensor fusion with NVIDIA DeepStream 9.0 and MV3DT to architect a unified multi-site tracking pipeline, improving cross-camera person tracking accuracy by **80%** and enabling **50+ concurrent camera streams** across locations at **24/7 uptime**.
- Integrated NVIDIA AutoMagicCalib with a LiDAR-derived floor plan contract to automate camera calibration, replacing fully manual projection matrix tuning and reducing calibration time per site by **90%**.
- Designed a fault-tolerant two-site fleet topology with a strict data plane / management plane separation using RTSP/SRT mesh backhaul and a Tailscale overlay, guaranteeing video processing continuity independent of operator network availability and supporting **6-8 cameras per location** on a single Blackwell GPU.